

ZHAW – Zurich University of Applied Sciences

MASTER THESIS

Track 1

Fitness Effects of Mutations to SARS-CoV-2 Proteins: A Sequence-Based Phylogenetic Inference Approach

by

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Abstract

This thesis investigates the fitness effects of mutations across SARS-CoV-2 proteins using a large-scale, sequence-based phylogenetic inference approach. Leveraging approximately 7 million SARS-CoV-2 genome sequences from the UShER mutation-annotated phylogenetic tree, the study computes expected versus actual mutation counts to derive amino acid fitness estimates. These computational estimates are systematically validated against experimental measurements from Deep Mutational Scanning (DMS) datasets, covering key viral proteins including the Spike receptor-binding domain (RBD), the full Spike protein, and the Main Protease (Mpro). The analysis reproduces and extends the Bloom & Neher (2023) pipeline, evaluates the Enrich2 weighting framework for improved replicate reliability, and identifies opportunities to extend validation to the Nucleocapsid (N) protein and to compare clade-specific fitness landscapes across Delta and Omicron variants.

1. Introduction

1.1 Background and Motivation

The COVID-19 pandemic, caused by SARS-CoV-2, has resulted in unprecedented global health challenges. As the virus evolved, numerous mutations emerged across its genome, some conferring advantages in transmissibility or immune escape, others being neutral or deleterious. Understanding the fitness landscape of these mutations, how each amino acid change affects the virus's ability to survive and transmit, is critical for anticipating future variants of concern, informing vaccine design, and identifying antiviral drug targets.

Traditional approaches to measuring mutation fitness rely on experimental techniques such as Deep Mutational Scanning (DMS), in which all possible single amino acid substitutions in a protein are assayed simultaneously. While DMS provides high-resolution experimental data, it is resource-intensive, limited to specific proteins, and subject to experimental noise including PCR bias and sequencing variability. A complementary strategy is to infer fitness computationally from the evolutionary record embedded in millions of sequenced viral genomes.

1.2 Research Objectives

Primary objectives:

- To understand and reproduce the computational framework of Bloom & Neher (2023) for inferring amino acid fitness effects from large-scale SARS-CoV-2 phylogenetic data.
- To evaluate the reliability of Deep Mutational Scanning datasets through replicate correlation analysis and weighted scoring using the Enrich2 framework.
- To assess whether combining phylogenetic sequence data with DMS experimental results provides biologically meaningful fitness estimates.

- To explore gaps in the existing analysis and propose extensions, including validation on the Nucleocapsid (N) protein and clade-specific comparisons between Delta and Omicron variants.
- To evaluate whether a time-resolved phylogenetic tree (TimeTree) provides more accurate fitness estimates than the clade-based approach, particularly for accessory proteins.

1.3 Hypothesis

A time resolved phylogenetic tree will produce more accurate mutation fitness estimates than the clade-based approach of Bloom & Neher (2023), particularly for SARS-CoV-2 accessory proteins (ORF3a, ORF6, ORF7a, ORF8). These accessory proteins have the fewest observed mutations and are therefore most sensitive to temporal sampling bias.

2. Background and Literature Review

2.1 SARS-CoV-2 Genome Architecture

SARS-CoV-2 is a positive-sense, single-stranded RNA betacoronavirus approximately 30 kilobases in length. Its genome encodes two large polyproteins (ORF1a/ORF1ab) that are cleaved into 16 non-structural proteins (nsps), as well as four structural proteins Spike (S), Envelope (E), Membrane (M), and Nucleocapsid (N) and several accessory proteins (ORF3a, ORF6, ORF7a, ORF7b, ORF8, ORF10).

The Spike protein mediates host cell entry via the ACE2 receptor and is the primary target of neutralizing antibodies and vaccines. The Main Protease (Mpro, encoded by nsp5) is essential for viral polyprotein processing and is a leading antiviral drug target. The Nucleocapsid protein plays critical roles in genome packaging and has been shown to be under strong purifying selection.

2.2 Phylogenetic Inference of Mutation Fitness

The Bloom & Neher (2023) approach infers mutation fitness by comparing actual observed mutation counts in SARS-CoV-2 phylogenetic trees against the expected counts under a neutral evolutionary model. The key insight is that mutations under purifying selection will be observed less frequently than expected, while adaptive mutations appear more frequently. This approach, formalized as:

$$\text{fitness} \sim \log(\text{actual_count} / \text{expected_count})$$

allows genome-wide, unbiased estimation of fitness effects for every possible amino acid substitution across all viral proteins simultaneously, using millions of real-world sequences.

2.3 Deep Mutational Scanning

Deep Mutational Scanning (DMS) is an experimental technique that comprehensively measures the functional effects of all possible single amino acid mutations in a protein. In the context of SARS-CoV-2, DMS datasets have been generated for the Spike RBD (Starr et al.,

Dadonaite et al.), the full Spike protein, and Mpro (Flynn et al. 2022 & 2023, Iketani et al.). These provide experimental ground truth for validating computational fitness estimates.

2.4 Four-Fold Degenerate Sites and Mutational Rate Estimation

A key methodological innovation in the Bloom & Neher approach is the use of four-fold degenerate (4FD) codon positions to estimate neutral mutation rates. At 4FD sites, any nucleotide substitution at the third codon position results in a synonymous change i.e., the same amino acid meaning these mutations are selectively neutral. By computing mutation rates at 4FD sites stratified by clade, the model can predict how often any given non-synonymous mutation would be expected to occur under neutral drift, against which actual counts are compared.

2.5 Enrich2: Weighted Scoring for DMS Data

Enrich2 is a software tool that uses weighted linear regression and a random-effects model to score variants from DMS experiments more accurately than previous ratio-based methods. Its core principle is inverse-variance weighting: barcodes with higher read counts are given proportionally more weight because they provide more statistically reliable measurements. This approach is particularly valuable when replicate barcodes have highly variable coverage, which is common in DMS experiments.

Known limitations of the Enrich2 pipeline include:

- Assumption of constant selection pressure across the experiment.
- Wild-type normalization is not always biologically valid.
- Cross-condition comparisons are unreliable without careful normalization.
- Replicates sharing a common starting library may miss systematic artifacts.
- Only handles point mutations; insertions and deletions are not supported by default.

3. Datasets

3.1 Natural SARS-CoV-2 Sequence Dataset

The primary dataset consists of approximately 7 million SARS-CoV-2 genome sequences obtained from the UShER (Ultrafast Sample placement on Existing tRees) mutation-annotated phylogenetic tree. Multiple time-stamped snapshots were used to ensure comprehensive temporal coverage:

- results_gisaid_2023-03-29
- results_gisaid_2024-04-24
- results_public_2022-01-31
- results_public_2022-12-18
- results_public_2023-03-27
- results_public_2023-05-11
- results_public_2023-06-21
- results_public_2023-10-01

- results_public_2024-04-19
- results_public_2024-11-01

3.2 Key Dataset Files

3.2.1 mutation_counts/aggregated.csv

Records every mutation observed in the phylogenetic tree with the following key fields:

Field	Description	Example
protein	Viral protein name	ORF1ab
aa_mutation	Amino acid change	P5401L
nt_mutation	Nucleotide change	C16466T
codon_change	Codon-level change	CCA>CTA
count	Total independent occurrences	81
count_terminal	Occurrences in terminal branches	44
count_non_terminal	Occurrences in internal branches	37
mean_log_size	Mean log clade size	0.6555
four_fold_degenerate	Whether site is 4FD	False

3.2.2 synonymous_mut_rates/rates_by_clade.csv

Provides per-clade neutral mutation rates estimated from synonymous substitutions at 4FD sites:

```
fraction = count_of_mutation_type / total_mutation_count_in_clade
rate = fraction / parent_nucleotide_frequency
```

3.2.3 expected_vs_actual_mut_counts.csv

Combines mutation rates with branch lengths and sequence opportunities to compute the expected count of each mutation under neutrality, compared to the actually observed count. Mutations with exclude=True or masked_in_usher=True are filtered out to avoid low-quality sites.

3.2.4 aa_fitness/aa_fitness.csv

The final output of the Bloom & Neher pipeline, containing per-amino-acid fitness estimates:

```
fitness ~ log(actual_count / expected_count)
```

Entries with high expected_count (many chances to occur) but low actual_count receive strongly negative fitness scores, indicating purifying selection. A threshold of expected_count >= 20 is applied to ensure statistical reliability.

3.3 Deep Mutational Scanning Datasets

DMS Dataset	Protein	Publication
starr_rbd	Spike RBD	Starr et al.
dadonaite_ba1_spike	Full Spike (BA.1)	Dadonaite et al.
flynn_mpro_2022	Mpro (nsp5)	Flynn et al. 2022
flynn_mpro_2023	Mpro (nsp5)	Flynn et al. 2023
iketani_mpro	Mpro (nsp5)	Iketani et al.

3.4 Data Sources

- SARS2-mut-fitness repository: <https://github.com/jbloomlab/SARS2-mut-fitness>
- Spike DMS codon variants: https://github.com/dms-vep/SARS-CoV-2_Omicron_BA.1_spike_DMS_mAbs
- Mpro DMS (Flynn 2022): <https://cdn.elifesciences.org/articles/77433/elifesciences-77433-fig2-data1-v2.xlsx>

4. Methodology

4.1 Pipeline Overview

The methodology follows a sequence-based inference approach in which large-scale evolutionary data rather than laboratory experiments drives fitness estimation. The pipeline consists of four major stages: (1) mutation counting from the phylogenetic tree, (2) neutral rate calibration using 4FD sites, (3) computation of expected vs. actual mutation counts, and (4) derivation of amino acid fitness scores. Each step is designed to minimize technical noise and correct for systematic biases.

4.2 Neutral Mutation Rate

Synonymous mutation rates at four-fold degenerate codon sites are computed separately for each viral clade and nucleotide context. For example, a 4FD site is one at which any nucleotide change at the third codon position results in the same amino acid such as the GGN codons, all encoding Glycine. These sites are assumed to evolve under neutral drift, making them ideal calibrators for background mutation rates.

4.3 Expected Count Calculation

The expected count of a given non-synonymous mutation is computed by multiplying the relevant neutral mutation rate by the total evolutionary opportunity for that mutation to occur across all branches of the phylogenetic tree. This opportunity is weighted by clade size (via `mean_log_size`) and branch lengths.

$$\text{expected_count} = \text{neutral_rate} \times \text{branch_length_opportunities}$$

4.4 Fitness Score Derivation

The fitness score for each amino acid mutation is defined as the log ratio of observed to expected counts. A mutation seen much less than expected is strongly deleterious (purifying selection); a mutation seen more than expected is potentially beneficial (positive selection).

$$\text{fitness} = \log(\text{actual_count} / \text{expected_count})$$

A minimum threshold of `expected_count >= 20` is applied to ensure that fitness estimates are based on sufficient evolutionary signal. The effect of this threshold on correlation with DMS experimental data was evaluated:

Filter Threshold	n (Mpro)	Pearson r (Mpro)
<code>expected_count >= 5</code>	1841	0.505
<code>expected_count >= 10</code>	1837	0.505
<code>expected_count >= 20</code>	1279	0.562

4.5 DMS Data Reproduction Workflow

The DMS data reproduction and integration pipeline consists of the following steps:

1. Import required libraries (pandas, numpy, scipy, matplotlib, seaborn).
2. Load fitness estimates from `aamut_fitness_all.csv` and apply `expected_count >= 20` filter.
3. Load Spike DMS barcode data: `codon_variants.csv` (barcode-to-variant lookup) and `variant_counts/*.csv` (per-sample read counts).
4. Merge read counts with the barcode-variant lookup table.
5. Filter to single amino acid mutants and compute enrichment scores as $\log_2(\text{output}/\text{input})$ ratios.
6. Normalize DMS scores: wildtype = 0, stop codon = -1.
7. Parse mutation strings (e.g., 'N501Y' -> wildtype=N, site=501, mutant=Y) to enable merging.
8. Merge Spike DMS scores with Bloom & Neher fitness estimates.
9. Process Mpro DMS data (Flynn et al. 2022) from the eLife supplementary data.
10. Generate scatter plots of DMS functional scores vs. sequence-based fitness estimates.

4.6 Enrich2 Weighted Scoring

To improve on simple $\log_2$ -ratio averaging, the Enrich2 inverse-variance weighting approach was implemented. Each barcode's score is weighted by its count reliability:

$$\begin{aligned} \text{weight} &= 1 / \text{variance}(\log_2_score) \\ \text{weighted_score} &= \text{sum}(\text{weight} * \text{score}) / \text{sum}(\text{weight}) \end{aligned}$$

This approach gives proportionally more influence to barcodes with high read coverage and suppresses the contribution of noisy low-count barcodes. For example, a barcode with 100

input reads receives far more weight than one with only 2 reads, whose log2 ratio is dominated by sampling stochasticity.

4.7 Replicate Quality Assessment

To verify experimental reproducibility, VSVG control replicates (VSVG_control_1 vs. VSVG_control_2) were compared using Pearson correlation. Barcodes present in both replicates were identified and their log2-ratio scores were compared:

- Total barcodes in both VSVG replicates: 94,355
- VSVG replicate Pearson $r = 0.9592$

The high replicate correlation ($r = 0.96$) confirms that the Spike DMS experimental data is of high quality and that technical variability between replicates is minimal.

5. Results

5.1 Reproduction of Bloom & Neher Figure 4

The primary validation target is Figure 4 of Bloom & Neher (2023), which compares sequence-based fitness estimates against DMS experimental scores. The analysis pipeline was reconstructed to reproduce this comparison, focusing on two proteins for which DMS data was available: Spike (via Dadonaite et al.) and Mpro (via Flynn et al. 2022).

The results obtained from the reproduced pipeline are:

Protein	n (mutations)	Pearson r (Reproduced)	Pearson r (Paper)
Spike (Dadonaite BA.1)	1855	0.258	~0.664
Mpro (Flynn 2022)	1279	0.562	~0.528

The Mpro correlation ($r=0.562$) closely reproduces the paper's reported value ($r\sim0.528$), validating the computational pipeline. The Spike correlation ($r=0.258$) is lower than the paper's reported $r\sim0.664$, likely due to differences in DMS normalization, filtering parameters, or the specific DMS condition used. Investigating this discrepancy through EnrichR2-based scoring is the subject of ongoing analysis.

5.2 VSVG Replicate Correlation

The VSVG control replicate analysis revealed excellent agreement between experimental replicates (Pearson $r=0.9592$ across 94,355 shared barcodes). Each data point in the replicate scatter plot represents one barcode's log2-ratio score computed using VSVG_control_1 as input (x-axis) vs. VSVG_control_2 as input (y-axis). Points on or near the diagonal indicate consistent measurements across replicates, confirming experimental reliability.

5.3 EnrichR2 vs. Simple Log2 Comparison

The EnrichR2 inverse-variance weighted scoring approach was compared against simple log₂-ratio averaging. Preliminary results indicate that EnrichR2 improves signal-to-noise ratio, particularly for mutations covered by low-count barcodes. The full comparison of Spike replicate correlation using both methods is being finalized.

5.4 Expected Count Threshold Analysis

The choice of expected_count filter has a meaningful impact on correlation quality. At threshold ≥ 20 , Mpro achieves $r=0.562$ ($n=1279$), superior to lower thresholds despite the reduced sample size, confirming that statistical reliability of fitness estimates outweighs coverage breadth.

6. Discussion

6.1 Strengths of the Sequence-Based Approach

The Bloom & Neher sequence-based fitness inference approach offers several important advantages over purely experimental strategies. It leverages the cumulative evolutionary signal from millions of real SARS-CoV-2 infections worldwide, providing a fitness measurement that reflects actual host-virus dynamics rather than in vitro selection conditions. It is also unbiased with respect to protein function — all proteins in the viral genome are simultaneously analyzed using the same framework.

6.2 Identified Gaps and Proposed Extensions

6.2.1 Extension to Nucleocapsid (N) Protein

The original Bloom & Neher paper validated fitness estimates experimentally for only two proteins (Spike and Mpro). A natural extension is to validate estimates for the Nucleocapsid (N) protein, which is essential for viral replication, is a vaccine and drug target, and is shown to be under strong purifying selection in the sequence-based analysis. DMS data for the N protein is publicly available, enabling direct validation.

6.2.2 Delta vs. Omicron Clade-Specific Fitness

The original paper aggregated all clades together, potentially masking important clade-specific fitness variation. Delta and Omicron are immunologically and structurally very different, and the same mutation can have opposite fitness consequences depending on the variant background. For example, mutation K444N in Spike was not beneficial in Delta but conferred strong immune escape benefits in Omicron BA.5. A clade-stratified analysis would recover this biologically important information.

6.2.3 Time-Resolved Phylogenetic Tree Analysis

The existing clade-based approach groups sequences by variant lineage rather than by time, which can introduce sampling bias for proteins with few observed mutations. A time-resolved phylogenetic tree (TimeTree approach) would allow fitness estimates to account for temporal dynamics of the pandemic, and may produce more accurate estimates for accessory proteins (ORF3a, ORF6, ORF7a, ORF8) that have limited mutation counts.

6.3 Limitations

- The fitness estimates are derived from natural sequence evolution and may conflate selection pressure differences across host immune backgrounds, tissue tropism, and geographic sampling.
- The expected count model assumes independence of mutations; epistatic interactions between co-occurring mutations are not accounted for.
- DMS experiments are conducted in vitro and may not capture all selective pressures operating in vivo.
- The Enrich2 pipeline assumes constant selection pressure and does not handle insertion/deletion variants.

7. Conclusion

This thesis successfully reproduces the core computational pipeline of Bloom & Neher (2023) for inferring the fitness effects of SARS-CoV-2 mutations from large-scale phylogenetic sequence data. The Mpro correlation with DMS experimental scores ($r=0.562$) closely matches the published value, validating the implementation. The Spike correlation ($r=0.258$) remains below the published value and motivates further investigation into DMS normalization strategies, including the EnrichR2 inverse-variance weighting approach.

The analysis demonstrates that sequence-based fitness inference is a powerful, scalable complement to experimental DMS approaches, capable of covering all viral proteins simultaneously using real-world evolutionary data. The high VSVG replicate correlation ($r=0.96$) confirms the quality of the experimental DMS data used for validation.

Key proposed extensions validation on the N protein, clade-specific Delta vs. Omicron comparisons, and time-resolved phylogenetic analysis represent meaningful opportunities to advance the understanding of SARS-CoV-2 evolutionary dynamics and to test the central hypothesis regarding temporal sampling bias in accessory protein fitness estimation.

References

- Bloom, J.D., & Neher, R.A. (2023). Fitness effects of mutations to SARS-CoV-2 proteins. *Virus Evolution*, 9(2), vead055.
- Dadonaite, B., et al. (2023). A pseudovirus system enables deep mutational scanning of all SARS-CoV-2 Omicron BA.1 spike mutations. *Cell*, 186(6), 1263-1278.
- Flynn, J.M., et al. (2022). Comprehensive fitness landscape of SARS-CoV-2 Mpro reveals insights into viral resistance mechanisms. *eLife*, 11, e77433.
- Starr, T.N., et al. (2020). Deep mutational scanning of SARS-CoV-2 receptor binding domain reveals constraints on folding and ACE2 binding. *Cell*, 182(5), 1295-1310.
- Rubin, A.F., et al. (2017). A statistical framework for analyzing deep mutational scanning data. *Genome Biology*, 18(1), 150. [Enrich2]
- Turakhia, Y., et al. (2021). Ultrafast Sample placement on Existing tRees (USHER) enables real-time phylogenetics for the SARS-CoV-2 pandemic. *Nature Genetics*, 53(6), 809-816.